ECA02 – DIGITAL CIRCUITS

SYSTEM MODELLING ASSIGNMENT

CO3: Design and Evaluation of Synchronous and Asynchronous Sequential Circuits

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Introduction

Sequential circuits are digital circuits whose outputs depend on the present inputs and the previous state (stored information). They use memory elements such as flip-flops. This assignment studies flip-flops, synchronous counters, asynchronous counters, finite state machines, simulation and performance.

Industrial example used: an automated manufacturing facility that moves materials using a conveyor, checks a machine, counts completed items and stops safely during an emergency.

Design assumptions: The assignment sheet does not specify an exact counting sequence or a particular simulator. Therefore, a 3-bit MOD-6 counter (counts 0 to 5 and repeats) is selected as an example industrial batch counter. A 3-bit ripple counter is used for asynchronous analysis. The FSM is a simple conveyor material-handling controller. These are clearly stated example designs.

Part (a) – Flip-Flop Modelling and State Analysis

1. Meaning of a flip-flop

A flip-flop is a basic memory circuit. It stores one binary bit: 0 or 1. Its output is called Q, and the opposite output is Q■. Most synchronous flip-flops change state at a clock edge.

2. SR flip-flop

S means Set and R means Reset. S=1 sets Q to 1; R=1 resets Q to 0. When S=R=0, the previous state is retained. For the active-high SR flip-flop, S=R=1 is invalid.

S	R	Q(next)	Operation
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	Invalid	Not allowed

Characteristic equation: Q(next) = S + R■Q, valid only when S and R are not both 1.

3. JK flip-flop

The JK flip-flop improves the SR flip-flop. When J=K=1, it toggles the output instead of entering an invalid condition.

J	K	Q(next)	Operation
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	Q■	Toggle

Characteristic equation: Q(next) = JQ■ + K■Q.

4. D flip-flop

D means Data or Delay. At the active clock edge, the value at D is copied to Q. It is commonly used to store data.

D	Q(next)	Operation
0	0	Store 0
1	1	Store 1

Characteristic equation: Q(next) = D.

5. T flip-flop

T means Toggle. If T=0, the output remains unchanged. If T=1, the output changes to its opposite value.

T	Q(next)	Operation
0	Q	Hold
1	Q■	Toggle

Characteristic equation: Q(next) = TQ■ + T■Q = Q ⊕ T.

6. Master-slave JK flip-flop

A master-slave JK flip-flop contains two stages: a master and a slave. The master accepts input during one clock level and the slave updates the output during the opposite level. This arrangement helps prevent repeated toggling (race-around) when J=K=1.

• J=K=0: output holds its value.

• J=1, K=0: output is set to 1.

• J=0, K=1: output is reset to 0.

• J=K=1: output toggles once for the clock operation.

State-transition idea: each flip-flop stores the present state and produces the next state according to its input and clock condition. The characteristic tables above describe these transitions.

Part (b) – Synchronous Sequential Circuit Design

1. Selected industrial sequence: MOD-6 batch counter

The counter tracks six stages/items: 0, 1, 2, 3, 4, 5 and then returns to 0. Three flip-flops are needed because $2^2 = 4$ is not enough, while $2^3 = 8$ can represent all six states. Let the present-state bits be Q2 Q1 Q0.

2. State diagram / sequence

000 → 001 → 010 → 011 → 100 → 101 → 000

Unused states: 110 and 111. In a practical implementation, these unused states should be directed to 000 so the counter can recover from an unexpected state.

3. State table

Present state Q2Q1Q0	Next state Q2■Q1■Q0■	Decimal count
000	001	0 → 1
001	010	1 → 2
010	011	2 → 3
011	100	3 → 4
100	101	4 → 5
101	000	5 → 0
110	000	Unused / recovery
111	000	Unused / recovery

4. Flip-flop selection and input equations

Use three D flip-flops because the D input directly becomes the next-state value. Therefore D2=Q2■, D1=Q1■ and D0=Q0■.

Flip-flop input	Next-state expression
D2	Q■2 Q1 Q0 + Q2 Q■1 Q■0
D1	Q■2 Q1 Q■0 + Q■2 Q■1 Q0
D0	Q■2 Q■0 + Q2 Q■1 Q0

The expressions are obtained by grouping the state-table rows where the corresponding next-state bit is 1. The unused states 110 and 111 are treated as recovery states with next state 000.

5. Working

• All three D flip-flops receive the same clock. This makes the counter synchronous.

• At each active clock edge, the three D inputs are loaded together.

• The displayed binary count advances from 000 to 101 and then returns to 000.

• A decoder or display circuit may be added to show the count to an operator.

Industrial use: count six items in one batch, six conveyor positions, or six repeated machine operations.

Part (c) – Asynchronous Sequential Circuit Design

1. Selected circuit: 3-bit asynchronous ripple counter

An asynchronous counter uses a clock only for the first flip-flop. The output of one flip-flop acts as the clock for the next flip-flop. The state change ripples through the stages, so the flip-flops do not change at exactly the same time.

2. Design and state transitions

Use three negative-edge-triggered T flip-flops with T=1 (or JK flip-flops with J=K=1). Apply the external clock to the first stage. Connect the first output to the clock of the second stage and the second output to the clock of the third stage. The counter has eight states.

Clock count	Q2 Q1 Q0	Decimal
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7
8	000	Repeats

3. Propagation delay

Propagation delay is the small time taken by a flip-flop to respond after its triggering input changes. In a 3-stage ripple counter, the change may pass through three stages. A simple worst-case estimate is:

t(total) ≈ $\mathbf{t_{pd1} + t_{pd2} + t_{pd3} \approx 3t_{pd}}$ (if all stages have approximately equal delay).

For example, if each flip-flop has a propagation delay of 20 ns, the approximate ripple delay is 3 × 20 ns = 60 ns. This is an illustrative value; the actual delay must be taken from the selected IC datasheet.

4. Race conditions and hazards

• Race condition: two or more state bits may change at slightly different times because of unequal propagation delays.

• Temporary intermediate states can appear during the ripple transition. A display or decoder may briefly show an incorrect value.

• A hazard is an unwanted short pulse caused by unequal signal paths. It may trigger another circuit if that circuit responds too quickly.

• Reduce problems by using proper timing margins, clean clock signals, suitable logic design and, where high speed is required, a synchronous counter.

Advantages: simple circuit and fewer clock connections. **Limitations:** accumulated delay and temporary output changes make it less suitable for high-speed control.

Part (d) – Finite State Machine Modelling

1. Material-handling controller

A finite state machine (FSM) is a controller that moves between a fixed number of states. The next state depends on the present state and the input. The controller below operates a conveyor that carries material to a machine.

2. Inputs and outputs

Symbol	Meaning
START	Start push button
ITEM	Item detected by sensor
DONE	Machine operation completed
ESTOP	Emergency-stop input; active when 1
MOTOR	Conveyor motor command
PROCESS	Machine processing command
ALARM	Warning indicator

3. Moore state model

A Moore FSM produces outputs based on the present state. This makes the output behavior easy to understand and verify.

State	Meaning	MOTOR	PROCESS	ALARM
S0 – IDLE	Waiting for start	0	0	0
S1 – MOVE	Move item on conveyor	1	0	0
S2 – PROCESS	Machine processes item	0	1	0
S3 – COMPLETE	Operation completed	0	0	0
S4 – EMERGENCY	Emergency stop active	0	0	1

4. State transitions

Present state	Input condition	Next state
S0 – IDLE	START=0	S0
S0 – IDLE	START=1 and ESTOP=0	S1
S1 – MOVE	ITEM=0 and ESTOP=0	S1
S1 – MOVE	ITEM=1 and ESTOP=0	S2
S2 – PROCESS	DONE=0 and ESTOP=0	S2
S2 – PROCESS	DONE=1 and ESTOP=0	S3
S3 – COMPLETE	Next cycle / START=1	S1
Any normal state	ESTOP=1	S4
S4 – EMERGENCY	ESTOP=1	S4
S4 – EMERGENCY	ESTOP=0 and RESET=1	S0

5. Emergency-stop response and safety

• Emergency input has priority over normal operation.

• When ESTOP=1, the controller enters S4 and commands MOTOR=0 and PROCESS=0; ALARM=1.

• The controller remains in emergency state until the emergency condition is cleared and a reset is given.

• For a real industrial machine, emergency-stop hardware must independently remove hazardous power. A classroom FSM is only a functional model and must not replace certified safety circuitry.

State sequence: IDLE → MOVE → PROCESS → COMPLETE → MOVE (next item). From any normal state, ESTOP=1 sends the controller to EMERGENCY.

Part (e) – Simulation and Functional Verification

1. Suggested simulation tool

The circuits can be implemented in Logisim Evolution, Proteus or another digital logic simulator. The following procedure is tool-independent.

2. MOD-6 synchronous counter test

• Place three D flip-flops and connect their clock pins to one common clock.

• Build the D2, D1 and D0 logic using AND, OR and NOT gates according to the equations in Part (b).

• Add three LEDs or a logic probe to observe Q2, Q1 and Q0.

• Apply a clock pulse and record the output after each active edge.

• Check that the sequence is 000, 001, 010, 011, 100, 101, then 000.

Test	Expected output	Result to verify
Reset	000	All outputs are 0
1st clock	001	Count increases to 1
2nd clock	010	Count increases to 2
3rd clock	011	Count increases to 3
4th clock	100	Count increases to 4
5th clock	101	Count increases to 5

Test	Expected output	Result to verify
6th clock	000	Counter returns to 0
Recovery test from 110/111	000	Unused states recover

3. Asynchronous counter test

• Place three T flip-flops and set T=1 for each stage (or JK with J=K=1).

• Connect the external clock to stage 0 and cascade the outputs as described in Part (c).

• Observe each output using logic probes or a timing diagram.

• Verify the binary sequence 000 to 111 and return to 000.

• Zoom into transitions to observe that later stages change after earlier stages because of propagation delay.

4. FSM test cases

Test	Input / condition	Expected response
1	Power-up / no START	IDLE; all outputs OFF
2	START=1, ESTOP=0	MOVE; motor ON
3	ITEM=1, ESTOP=0	PROCESS; process output ON
4	DONE=1, ESTOP=0	COMPLETE; motor and process OFF
5	ESTOP=1 in MOVE	EMERGENCY; motor/process OFF, alarm ON
6	ESTOP=1 in PROCESS	EMERGENCY; motor/process OFF, alarm ON
7	ESTOP cleared, RESET=1	Return to IDLE

Verification result: The designs are functionally correct when the observed outputs match the expected sequences and emergency input always forces the controller into the emergency state. Actual simulation screenshots should be added after running the circuits in the selected software; they are not claimed as completed here.

Part (f) – Performance Evaluation

1. Comparison

Parameter	Synchronous MOD-6 counter	Asynchronous 3-bit counter
Clock	Common clock to all flip-flops	Clock ripples from one stage to next
Timing	Flip-flops update together at clock edge	Stages update at different times
Delay	Mainly flip-flop and combinational logic delay	Accumulated stage propagation delay
Output glitches	Lower chance of ripple-related intermediate states	Temporary intermediate states may occur
Complexity	Needs next-state logic	Simple connections
Best use	Reliable, coordinated industrial counting	Simple low-speed counting

2. Maximum operating frequency

The maximum clock frequency depends on the flip-flop delay, setup time and combinational logic delay. For a synchronous design, a simplified timing condition is:

$$T_{clk} \geq t_{cq} + t_{logic} + t_{setup}$$

Therefore, $f_{max} \leq 1 / (t_{cq} + t_{logic} + t_{setup})$. The exact value cannot be calculated without the selected IC's datasheet timing values and the measured logic delay.

For a ripple counter, the clock period must also allow the ripple to settle through the stages. A simplified estimate is Tclk ≥ n × tpd, where n is the number of stages. This is only an approximate design check.

3. Reliability and architecture recommendation

• Use a synchronous counter for the industrial batch-counting function because all state bits update from the same clock and the output sequence is easier to coordinate.

• Use an asynchronous counter only for simple, low-speed counting where its ripple delay and temporary states are acceptable.

• Use an FSM for conveyor sequencing because it clearly separates IDLE, MOVE, PROCESS, COMPLETE and EMERGENCY operations.

• Use a separate, properly rated emergency-stop safety circuit in real equipment; software or a classroom logic model alone is not sufficient.

Overall evaluation: The synchronous MOD-6 counter is the preferred counting architecture for the selected manufacturing example. The FSM provides organized material movement and emergency response. The asynchronous counter demonstrates a simpler alternative but has greater timing limitations.

Conclusion

This assignment models five types of flip-flops and explains their state transitions. A synchronous MOD-6 counter, an asynchronous 3-bit ripple counter and a Moore material-handling FSM were designed. Test sequences and expected results were prepared for simulation. The comparison shows that synchronous circuits are more suitable for coordinated industrial control, while asynchronous counters are useful for simpler low-speed applications. Final timing limits must be confirmed using the chosen IC datasheets and actual simulation or hardware measurements.

References

• ECA02 Digital Circuits – CO3 System Modelling Assignment sheet supplied for this work.

• Digital logic design lecture notes and standard flip-flop characteristic-table concepts.

• Datasheets of the selected flip-flop ICs should be consulted for actual propagation delay, setup time and maximum clock frequency.

Note: This answer uses example design choices because the assignment brief does not provide a fixed industrial count sequence, a specific simulator or IC part numbers. Add your own simulator screenshots and measured timing values before final submission.